A Decision-Oriented Trend Radar for Sustainable Commercial Mobility

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1 Sector choice and scope

Automotive — zero-emission commercial vehicles (2031–2036). BLUE's 2026 R&D partnering profile specifically targets electric powertrains, battery durability, digital twins and data-driven circularity [1]. Its B4IA hydrogen-hybrid minibus project is scheduled to finish in December 2026. European demand has clear timing: manufacturers must reach 90% zero-emission shares in new urban-bus fleets by 2030 and 100% by 2035, while covered batteries require passports from February 2027 [2]. Rail is a credible alternative, but road transport combines these BLUE capabilities with a near-term funding route.

2 Method and eight selected signals

Twenty candidates were screened for BLUE fit, future relevance, R&D potential and source quality. **Industry TRL** rates a *specific, publicly evidenced application* on the European 1–9 scale [2]; an asterisk flags provisional evidence. The **R&D opportunity** is a distinct, unresolved next step; any starting maturity is a planning assumption, not a measured TRL. *Core* extends BLUE services, *adjacent* transfers them, and *transformative* may reshape how BLUE earns revenue. Radar rings recommend action: *Engage now*, *Build capability*, or *Monitor*. A weighted priority score ranks passports first in 99% of 5,000 weight variations (`research/`).

Trend	TRL	Class	Ring	Key evidence
1 MCS heavy-duty powertrains	8	Core	Engage	1 MW demo, 3 OEMs; MAN offers MCS
2 Battery health analytics	8	Adj.	Engage	Commercial e-bus SOH service since 2020
3 Vehicle/fleet twins	5	Core	Build	Minibus subsystem twin validated (MINDED)
4 Virtual homologation	6*	Core	Build	UN R171 allows assessed virtual tests
5 Level-4 shuttles	7	Adj.	Build	Driverless at Zurich Airport; operator on roads
6 Passports, circular data	7	Adj.	Engage	Passport law; Catena-X pilot
7 AI surrogates for CAE	8	Transf.	Build	Commercial CFD surrogate (May 2026)
8 Hydrogen hybrid buses	9	Core	Monitor	In service: Lorient, Bologna

Table 1: TRL at industry state of the art (* provisional). Each entry's sources and limitations are in `data/trends.json`.

What the radar says. Six of eight technologies are already at TRL 7–9 in industry; BLUE's R&D value lies in the unsolved layer above them: credible models, quantified uncertainty and lifecycle data.

3 Recommended R&D project: CircuTwin-HD

Uncertainty-aware lifecycle decisions for electric-bus traction inverters. A used inverter can be repaired, remanufactured, reused or recycled; simple rules on age, mileage and fault codes, used here as the baseline, cannot see the stress history that drives power-device wear. **Research question:** can a lifecycle twin combining measured load histories, a calibrated physics-of-failure model, explicitly quantified remaining-life uncertainty, and LCA/LCC support better decisions than a transparent rule-based baseline?

Novelty test. RUST2LIFE, CREDIT and RESURRECTION already connect twins, passports and remanufacturing [3], but their public descriptions do not specify this inverter, load-history and uncertainty-aware decision test; power-cycling research supplies a damage-model building block [4]. The gap is *testable*, not a claimed world first.

Pilot: (1) define traceable component identities and secure data access; (2) log authorised electrical and thermal loads on an operating e-bus fleet; (3) calibrate remaining-life estimates on power-cycling tests; (4) compare repair/reuse/remanufacture/recycling with LCA and cost; (5) assess recommendations blindly against expert decisions and returned-component inspections. Measures: prediction error and calibration, decision agreement, traceability, CO_2e. BLUE could lead simulation, twin architecture and validation, with an OEM, operator, university and remanufacturer; cheap stage gates with stop criteria come first (`research/FEASIBILITY.md`).

Funding fit. HORIZON-CL5-2027-03-D5-03, Area B, calls for component track-and-trace, measured operating loads, API-based data exchange and digital-twin passports with LCA [5]. The Work Programme lists a lump-sum **Innovation Action**: EUR 24m total, EUR 5–7m *per consortium project*, open 15 December 2026 to 14 April 2027 [5]. It is a consortium bid, not a BLUE grant; novelty, data rights and eligibility must be checked. Fallback: Italy's *Accordi per l'innovazione*, which funds B4IA.

4 Keeping the radar alive

An R&D owner runs quarterly reviews plus checks triggered by project ends, the Horizon call and passport milestones. One JSON file generates the dashboard, CSV and report table. A GitHub workflow checks source/data consistency monthly and opens a review issue for unacknowledged milestones. Reclassify a trend only with a dated source, justification and confidence; Git history keeps the audit trail. A trend without new evidence for two reviews moves to Monitor or is retired.

References

[1] BLUE Engineering, Cluster 5 partnering profile, Jul 2026.

[2] EU Regulations 2024/1610 and 2023/1542 (Art. 77); European Innovation Council, TRL definitions.

[3] CORDIS: RUST2LIFE, CREDIT, RESURRECTION.

[4] Kim, Lee, Han, IGBT physics-of-failure damage metric, PHME 2026.

[5] European Commission: CORDIS D5-03 scope (Area B); Work Programme 2026–2027, Cluster 5.